

- 3 (a) Explain what is meant by *internal energy* of a gas.

.....

 [1]

- (b) The pressure p of an ideal gas of density ρ is related to the mean-square speed of its molecules by the expression

$$p = \frac{1}{3} \rho \langle c^2 \rangle$$

Show that the average kinetic energy of a molecule of an ideal gas is proportional to its thermodynamic temperature T .

[2]

- (c) An engine operates by using 0.024 mol of an ideal gas. The gas undergoes a cycle of changes as shown in Fig. 3.1.

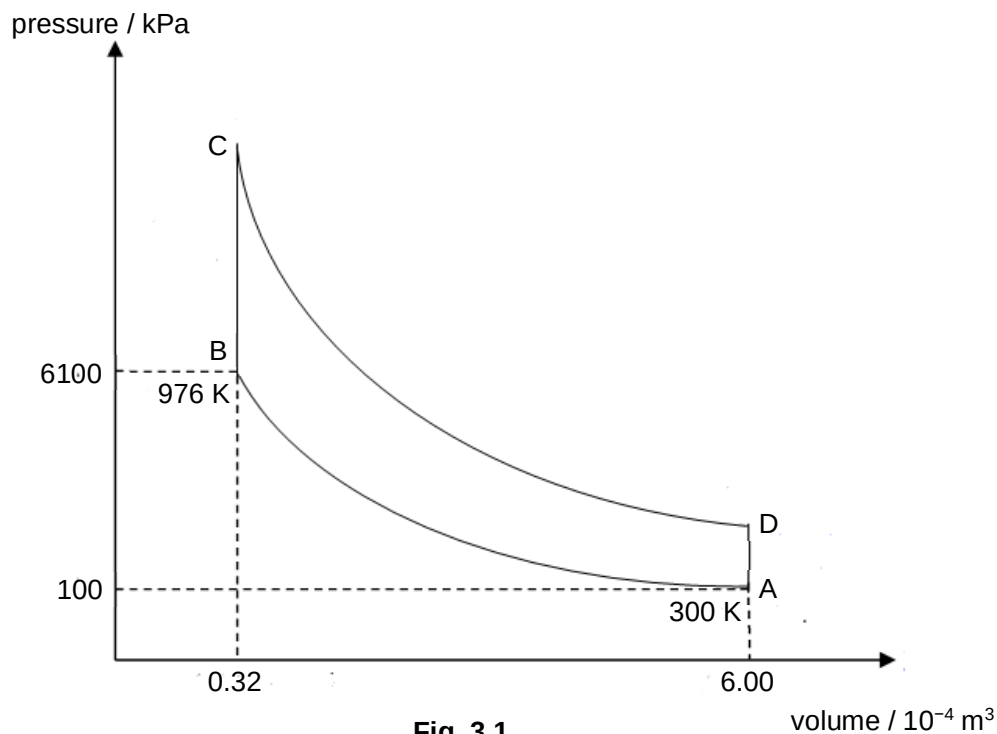


Fig. 3.1

The temperature of the gas at A and B are 300 K and 976 K respectively.

- (i) Given that the gas gains 250 J of heat during a heating process at constant volume from B to C, determine

1. the temperature at C,

temperature = K [3]

2. the pressure at C.

pressure = Pa [1]

- (ii) Fig. 3.2 shows some of the values of changes in Fig. 3.1.

Complete Fig. 3.2.

change	heat supplied to gas / J	work done by gas / J	increase in internal energy / J
A to B	0	-200	
B to C	250		
C to D	0		
D to A	-100		

Fig. 3.2

[3]